

A faithful singleton model forces simplicity

`MathlibAnnex.Analysis.CStarAlgebra.Representation.isSimpleCStarAlgebra_of_singleton_of_injective`

theorem

Uses irreducible representations annihilating proper ideals to rule out nonzero proper closed ideals.

Statement

Let A be a nonzero unital complex C^* -algebra, and let H be a complex Hilbert space. Let $\pi : A \rightarrow \mathcal{B}(H)$ be a unital $*$ -representation. Assume that π is injective and is a singleton irreducible model among unital representations. Then every norm-closed two-sided ideal I of A is either $\{0\}$ or A .

Assumptions

The singleton condition includes irreducibility on a nonzero space and unitary equivalence with every nonzero irreducible unital representation in the class of all complex Hilbert-space representations quantified in the definition. Injectivity is an explicit, separate hypothesis here. Neither A nor H is assumed separable.

Conclusion

A is simple in the sense of nontriviality and absence of nonzero proper norm-closed two-sided ideals.

Proof route

An irreducible GNS representation kills a proper closed ideal. Transfer its zero action back through the singleton unitary, then use faithfulness.

▼ Proof steps

1. Fix a closed two-sided ideal I . If $I = A$ there is nothing to prove. Otherwise apply A pure GNS representation annihilating a proper closed ideal with this proper closed I in the nonzero ordered unital algebra A . It supplies a pure state φ and an irreducible GNS representation $\rho : A \rightarrow \mathcal{B}(H_\varphi)$ with $\rho(a) = 0$ for every $a \in I$. Its cyclic vector has norm one, so $H_\varphi \neq 0$.

2. The singleton assumption, applied to this very ρ , gives a unitary $U : H \rightarrow H_\varphi$ such that

$$U(\pi(a)x) = \rho(a)(Ux) \quad (a \in A, x \in H).$$

If $a \in I$, the right side is zero. Injectivity of U therefore gives $\pi(a)x = 0$ for every x , hence $\pi(a) = 0$.

3. Injectivity of π and $\pi(0) = 0$ yield $a = 0$. Thus $I \subseteq \{0\}$ and $I = \{0\}$. Together with the first case and the given $A \neq 0$, this is precisely the definition of simplicity.

Main citations

- Exact theorem and proof
- A pure GNS representation annihilating a proper closed ideal
- Definition of closed-ideal simplicity

Lean source signature (exact)

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theorem isSimpleCStarAlgebra_of_singleton_of_injective [Nontrivial A]
  (pi : Representation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)
  (hinj : Function.Injective pi) : IsSimpleCStarAlgebra A
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
pi : Representation A H	The unital complex $*$ -representation π on the complete Hilbert space H ; the surrounding context also equips A with the usual C^* -algebra order.
hsingle : IsSingletonIrreducibleModel.{u, v, u} pi	π is irreducible on a nonzero H , and every nonzero irreducible unital comparison representation is unitarily equivalent to it.
hinj : Function.Injective pi	Equality of represented operators $\pi(a) = \pi(b)$ implies $a = b$. This is a hypothesis of this theorem.
IsSimpleCStarAlgebra A	$A \neq 0$, and for every norm-closed two-sided ideal I of A , either $I = \{0\}$ or $I = A$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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